Approximating Meta Value of Champions and Champion Interactions in Professional *League of Legends*

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Abstract

We describe a method for learning interpretable, role-aware *value functions* for the champion draft phase of competitive *League of Legends*. Each draft action (pick or ban) is scored by an additive decomposition into a champion-role strength term, a low-rank antisymmetric matchup term against enemy picks, and a low-rank symmetric synergy term with ally picks. The vocabulary unit is the (CHAMPION, ROLE) token, which lets the model represent the common case in which a champion behaves very differently across the five canonical roles. To handle the rapidly shifting professional meta, we partition the season into chronological game-count windows, fit independent models on each, combine the resulting matchup and synergy structures with an exponential-decay weighting, and finally refit only the per-token strength term (and a small set of additive biases) on the most recent patches with the interaction structure held fixed. The resulting model produces calibrated draft-action distributions, dense interpretable matchup and synergy matrices, and a strength signal that tracks short-horizon patch changes without losing access to longer-term interaction data.

1 Introduction

Competitive *League of Legends* matches begin with an alternating pick/ban phase in which two teams of five players draft champions from a roster of roughly 170 unique characters. The draft determines matchups, lane assignments, team composition archetypes, and a substantial share of expected win probability before any in-game action occurs. For a professional coaching staff, an accurate read on the current *meta* – which champions and which compositions the rest of the competitive field considers strong, and in what draft order they are willing to commit to them – is a core input to scouting, preparation, and in-series adaptation. Modeling the draft is therefore of interest both to teams preparing for opponents and to analysts producing post-hoc explanations of match outcomes.

Two features of the professional setting make this modeling problem hard. First, the data is sparse: a full season of top-flight matches produces on the order of thousands of games, which is small relative to the combinatorics of a roster of roughly 170 champions, five roles, and twenty ordered draft slots. Second, the obvious source of additional volume – solo-queue ranked ladder data – is not directly trusted by professional staff: the solo-queue meta is shaped by very different incentives (individual performance over team coordination, blind pick rather than negotiated draft, no bans against specific opponents), and champion strengths in that environment routinely disagree with the professional consensus. The dominant publicly available draft-evaluation tools (for example DraftGap and similar solo-queue oriented systems) inherit this mismatch and additionally tend to treat champions as roleless atoms or assume a single canonical role per champion.

Within professional analytics the standard summaries are per-champion win rate and pick/ban *presence* (the fraction of games in which a champion was either picked or banned by some team). These statistics are easy to compute and easy to read, but they discard essentially all of the *context* in which a champion appeared: who they were picked into, what teammates they were picked alongside, how early in the draft a team was willing to commit to them, and whether opponents respected them enough to spend an early ban. A champion picked last-rotation as a flex into a known counter and a champion first-picked blueside on patch day register identically in a presence table, yet a coach reads them very differently. We want a more robust way of recovering what coaches actually think is strong: one that uses the *order* of picks and bans, not just their presence, and that conditions each action on the partial draft state at the moment of that decision. Professional play has qualitatively different statistics from solo queue beyond just sample size: draft strategy is highly intentional, and the meta turns over frequently with patches. These properties reward methods that share information across roles within a champion while still being responsive to patch-level shifts.

We make three design choices to address this setting:

1. **Per-role tokenization.** We treat each (CHAMPION, ROLE) pair seen with sufficient frequency as a distinct token. Multi-role champions therefore receive distinct strength and interaction parameters for each role they play, while single-role champions collapse to a single token. Optional parameter sharing pulls all role-tokens of a champion toward a common base.

2. **Low-rank additive interactions with the correct symmetries.** Matchup is parameterized as the antisymmetric inner product of two low-rank factor matrices, so by construction swapping the two participants negates the term. Synergy is parameterized as a Gram matrix of a single low-rank factor and is therefore symmetric. Both compose linearly with the per-token strength, so a single forward pass produces logits for every legal action.

3. **Game-count windowing with structure freezing.** The interaction terms are estimated by combining per-window fits with exponential decay over windows; the per-token strength term is then refit on the most recent patches against this frozen interaction structure. This separates a stable estimate of *how champions interact* from a fast-moving estimate of *which champions are currently strong.*

The remainder of the paper formalizes the data and notation (Section 2), defines the model (Section 3), describes the training objective and masking (Section 4), introduces the windowed-meta pipeline (Section 5), summarizes ablations qualitatively (Section 6), and discusses limitations and extensions (Section 7).

2 Data and Notation

We work with publicly available professional match data covering a single *League of Legends* season. Each match consists of one or more games; each game produces a sequence of 20 ordered draft actions (10 bans and 10 picks) alternating between the two sides in the standard tournament order. For each pick we additionally recover the in-game role assignment (top, jungle, mid, bot, support) from the post-game roster, using a separate preprocessing step.

We write $\mathcal{C}$ for the set of champions and $\mathcal{R} = \{\text{TOP}, \text{JNG}, \text{MID}, \text{BOT}, \text{SUP}\}$ for the set of canonical roles. The *(champion, role) token vocabulary* is

$$\mathcal{T} = \big\{(c, r) \in \mathcal{C} \times \mathcal{R} \; : \; n(c, r) \geq \tau\big\},$$

where $n(c, r)$ counts the number of in-window pick observations of champion c at role r, and τ is a tunable minimum-plays threshold. We use $T = |\mathcal{T}|$ and $C = |\mathcal{C}|$. Each token $t \in \mathcal{T}$ has an associated champion $c(t)$ and role $r(t)$.

A draft state $\mathbf{x}$ at the moment of an action consists of the acting side, the side's $\mathcal{A}(\mathbf{x}) \subseteq \mathcal{T}$ of allied picks (each committed to a specific role), the opposing side's $\mathcal{E}(\mathbf{x})$, the set of used champions $\mathcal{U}(\mathbf{x}) \subseteq \mathcal{C}$ (picked by either side or banned), and the ordinal slot index $k(\mathbf{x}) \in \{0, \ldots, 19\}$.

Augmentation. Within a same-side pick block (for example the second and third picks of red side in the first phase, which the two players resolve simultaneously) the canonical sequence ordering is not informative. We emit both orderings of every same-team block as training rows, doubling the apparent number of decisions per game without introducing label noise. Group-shuffle splits on the match identifier prevent the two orderings from leaking across the train/validation/test partition.

Splits. All splits are performed at the level of the match identifier (best-of-N series) using a grouped shuffle. This prevents games from a single series, which share rosters and strategy, from appearing on both sides of the split.

3 Model

3.1 Per-token parameters

For each token $t \in \mathcal{T}$ the model maintains:

- a strength scalar $s_t \in \mathbb{R}$;
- two matchup factors $A_t, B_t \in \mathbb{R}^{k_m}$;
- one synergy factor $S_t \in \mathbb{R}^{k_s}$.

3.2 Pairwise interactions

The matchup contribution of token t against an enemy token e is

$$M(t, e) \;=\; \langle A_t, B_e \rangle \;-\; \langle B_t, A_e \rangle. \tag{1}$$

By construction $M(e, t) = -M(t, e)$ for every pair, so the matchup matrix $M \in \mathbb{R}^{T \times T}$ assembled from (1) is exactly antisymmetric. This encodes the physical assumption that if t counters e by some margin, e counters t by the opposite margin. The bilinear-factor parameterization is the canonical low-rank relaxation of an arbitrary antisymmetric matrix $M = AB^\top - BA^\top$.

The synergy contribution of token t with an ally token a is

$$\Sigma(t, a) \;=\; \langle S_t, S_a \rangle. \tag{2}$$

The assembled $\Sigma \in \mathbb{R}^{T \times T}$ is the Gram matrix of S and is therefore symmetric and positive semidefinite. We interpret S_t as a low-dimensional embedding of the role t plays in a team composition; tokens with aligned embeddings reinforce each other.

3.3 Pick value

The score the acting side assigns to picking token t in state $\mathbf{x}$ is

$$v^{\text{pick}}(t \mid \mathbf{x}) \;=\; s_t \;+ \sum_{e \in \mathcal{E}(\mathbf{x})} M(t,e) \;+ \sum_{a \in \mathcal{A}(\mathbf{x})} \Sigma(t,a). \tag{3}$$

Because the sums in (3) factor through $M = AB^\top - BA^\top$ and $\Sigma = SS^\top$, computing the vector of pick logits for every token in a batch reduces to three dense matrix products against multi-hot encodings of $\mathcal{A}(\mathbf{x})$ and $\mathcal{E}(\mathbf{x})$.

3.4 Ban value

A ban removes a champion from the enemy's available pool. We model the ban target as the champion the enemy would most plausibly pick next, were the ban not used. Concretely, we first compute a token-level ban score with the perspective swapped relative to (3):

$$v^{\text{ban}}_t(t \mid \mathbf{x}) \;=\; s_t \;+ \sum_{a \in \mathcal{A}(\mathbf{x})} M(t,a) \;+ \sum_{e \in \mathcal{E}(\mathbf{x})} \Sigma(t,e). \tag{4}$$

Here t ranges over tokens whose role is still open on the enemy side and whose champion is not yet used. The per-champion ban logit collapses across the champion's legal role-tokens by the log-sum-exp

$$v^{\text{ban}}(c \mid \mathbf{x}) \;=\; \operatorname*{logsumexp}_{r \in \mathcal{R}(c \mid \mathbf{x})} v^{\text{ban}}_t\big((c,r) \mid \mathbf{x}\big), \tag{5}$$

where $\mathcal{R}(c \mid \mathbf{x}) \subseteq \mathcal{R}$ is the set of roles that the champion has any in-vocabulary token at and that remain open on the enemy team. Equation (5) is the log-marginal-likelihood over an unobserved role assignment, treating each role-token's score as the log of an unnormalized propensity.

3.5 Additive biases

We optionally augment the per-token score with three additive bias families:

- a champion bias $\beta^{\text{champ}}_{c(t)} \in \mathbb{R}$, shared across all role-tokens of the same champion;
- a side bias $\beta^{\text{side}}_{t,\sigma} \in \mathbb{R}$ for $\sigma \in \{\text{blue}, \text{red}\}$, capturing systematic side advantages;
- a slot bias $\beta^{\text{slot}}_{t,k} \in \mathbb{R}$ for the ordinal draft slot $k(\mathbf{x})$, capturing position-in-draft effects (for example, blueside-first picks versus last picks).

When enabled, these biases are added to both v^{pick} and the token-level v^{ban}_t before any logit masking.

3.6 Champion-base + role-residual sharing

In an alternative parameterization, each per-token parameter is expressed as the sum of a champion-level base and a token-level residual,

$$\theta_t \;=\; \theta^{(\mathcal{C})}_{c(t)} \;+\; \tilde{\theta}_t, \qquad \theta \in \{s, A, B, S\}, \tag{6}$$

with an additional ℓ_2 penalty $\lambda_{\text{res}} \|\tilde{\theta}\|_2^2$ on the residuals. With λ_{res} large the model collapses to a roleless champion-only model; with λ_{res} small it recovers the unconstrained per-token model. This decomposition is most useful for multi-role champions with uneven role sample sizes: the rare-role token inherits the common-role token's structure and only deviates where data warrants. At export time the residuals and bases are materialized back to the flat per-token layout used in (3)–(5), so downstream consumers are unchanged.

4 Training

4.1 Legality masks

The model only assigns probability mass to legal actions in each state.

For pick actions, the legal-token mask is

$$\text{legal}^{\text{pick}}(t \mid \mathbf{x}) \;=\; \big[r(t) \in \mathcal{R}^{\mathcal{A}}_{\text{open}}(\mathbf{x})\big] \cdot \big[c(t) \notin \mathcal{U}(\mathbf{x})\big], \tag{7}$$

where $\mathcal{R}^{\mathcal{A}}_{\text{open}}(\mathbf{x})$ is the set of roles not yet committed by the acting side's prior picks. Illegal tokens receive a logit of $-\infty$ before the softmax.

For ban actions, the same construction is applied to the enemy's open roles, and the per-champion log-sum-exp (5) is restricted to that champion's legal role-tokens.

4.2 Objective

We minimize a masked cross-entropy on the observed action. For a pick row with target token $t^\star$,

$$\ell^{\text{pick}}(\mathbf{x}, t^\star) \;=\; -\log \frac{\exp\big(v^{\text{pick}}(t^\star \mid \mathbf{x})\big)}{\sum_{t:\text{legal}^{\text{pick}}(t\mid\mathbf{x})} \exp\big(v^{\text{pick}}(t \mid \mathbf{x})\big)}. \tag{8}$$

For a ban row, the labeled champion $c^\star$ does not name a role. Multi-role champions therefore raise an attribution question: which of their role-tokens absorbed the ban credit? We supervise with a *soft* target distribution over the champion's legal role-tokens proportional to their in-vocabulary pick frequency,

$$q_t(c^\star \mid \mathbf{x}) \;\propto\; n(c^\star, r(t)) \cdot \big[t \in \mathcal{R}(c^\star \mid \mathbf{x})\big], \tag{9}$$

renormalized to sum to one. The ban loss is the cross-entropy of q against the token-level softmax of v^{ban}_t:

$$\ell^{\text{ban}}(\mathbf{x}, c^\star) \;=\; -\sum_t q_t(c^\star \mid \mathbf{x}) \log \frac{\exp\big(v^{\text{ban}}_t(t \mid \mathbf{x})\big)}{\sum_{t':\text{legal}^{\text{ban}}(t'\mid\mathbf{x})} \exp\big(v^{\text{ban}}_t(t' \mid \mathbf{x})\big)}. \tag{10}$$

This is equivalent to marginalizing the role attribution out of the observation under a frequency-weighted prior and recovers the per-champion log-sum-exp (5) as the implied champion-level score.

We additionally exclude low-signal ban observations from the loss (blueside bans 1–3 and either-side bans 4–5, which are formula-driven in current professional meta) to reduce label noise.

4.3 Regularization

The interaction factors receive a Frobenius penalty

$$\mathcal{R}_{\text{factor}} \;=\; \lambda_{\text{fac}}\big(\|A\|_F^2 + \|B\|_F^2 + \|S\|_F^2\big), \tag{11}$$

and, when the champion-base/role-residual parameterization of Section 3 is enabled, an additional residual penalty $\lambda_{\text{res}}\|\tilde{\theta}\|_2^2$ summed over $\theta \in \{s, A, B, S\}$. A separate optional strength penalty $\lambda_s\|s\|_2^2$ is available but typically not used.

4.4 Optimization

We optimize (8), (10), and the regularizers jointly with Adam, using early stopping on validation loss with a fixed patience. Pick and ban rows are interleaved freely within a batch; the legality masks ensure that the two losses share parameters but apply to disjoint sets of legal actions.

4.5 Auxiliary sequential loss (not used by default)

A Plackett–Luce auxiliary loss over the acting side's remaining picks in the same game was implemented and evaluated. The intent was to encourage the model to score not only the current pick but the entire future ordering of teammates consistently. In practice this loss did not improve held-out cross-entropy at the data sizes we considered and is kept in the code base but disabled in the default configuration.

5 Windowed Meta and Structure Freezing

The professional meta shifts on a timescale of weeks: patches change champion power, and team strategies adapt to new patches within days. A single fit on the full season therefore underweights recent observations. Conversely, fitting only on the most recent patch yields unstable interaction estimates because the per-window sample size is small relative to the $\binom{T}{2}$ entries of M and Σ.

We address this with a two-stage procedure that separates the estimation of *interaction structure* from the estimation of *current strength*.

5.1 Phase A: per-window fits

Patches are processed in chronological order and accumulated into *windows* by total distinct game count. A window closes when its cumulative game count first reaches a target $G^\star$; if the tail of the season produces a window much smaller than $G^\star$, it is folded into the previous window. Windows therefore consist of contiguous whole patches and have roughly comparable sample sizes.

A global token vocabulary is built once from the entire season's pick rows (Section 2), so that all windows share the same token index space. An independent model (Section 3) is then fit to each window using the objective of Section 4. Let $(s^{(w)}, A^{(w)}, B^{(w)}, S^{(w)})$ denote the fitted parameters of window w, with any optional biases denoted analogously.

5.2 Phase B: combining windows

For each window w we reconstitute the dense interaction matrices

$$M^{(w)} \;=\; A^{(w)} B^{(w)\top} \;-\; B^{(w)} A^{(w)\top}, \qquad \Sigma^{(w)} \;=\; S^{(w)} S^{(w)\top}. \tag{12}$$

We then combine across windows by a weighted average,

$$\bar{M} \;=\; \frac{\sum_w \omega_w M^{(w)}}{\sum_w \omega_w}, \qquad \bar{\Sigma} \;=\; \frac{\sum_w \omega_w \Sigma^{(w)}}{\sum_w \omega_w}, \tag{13}$$

with the same averaging applied to $s^{(w)}$ and any per-window biases. The weights are

$$\omega_w \;=\; n_w \cdot \exp\big(-\lambda(W-1-w)\big), \qquad \lambda \;=\; \frac{\ln 2}{H}, \tag{14}$$

where n_w is the number of distinct games in window w, W is the total number of windows, and H is the half-life expressed in windows. The age $W-1-w$ is zero for the most recent window, so that window receives full sample-size weight and earlier windows decay geometrically.

Averaging in the dense M, Σ space rather than at the level of the factors A, B, S avoids ambiguities arising from per-window rotational symmetry of the low-rank factors (the factors are only identifiable up to the orthogonal rotations that preserve their products).

5.3 Phase C: structure-frozen strength refit

Given the combined interaction structure $(\bar{M}, \bar{\Sigma})$, the final model holds these matrices fixed as buffers and re-estimates only s (and any enabled biases) on the most recent N patches. We call this model the *frozen-structure value model*; its forward pass replaces the matrix products of Section 3 with direct multiplications against $\bar{M}^\top$ and $\bar{\Sigma}^\top$, and its training loop is the same masked cross-entropy as Section 4 but with the gradient restricted to the strength and bias parameters.

This decomposition has a clean interpretation. Long-range relationships (matchup and synergy) accumulate evidence slowly and are estimated against the entire season with mild recency weighting. Short-range power (current per-token strength) is responsive to the current patch and is estimated cleanly against recent data without having to relearn the interaction structure from scratch.

6 Ablation Discussion

We summarize qualitatively which design choices improved held-out performance in our experiments. We omit numerical results in this methodology report.

What helped.

- Enabling all three additive bias families (champion, side, slot) consistently improved both pick and ban cross-entropy. The slot bias contributed the largest share, presumably because it captures the strongly position-dependent base rates of which champions are picked or banned at each draft slot.
- A modest increase in interaction rank improved fit without overfitting at this data scale; very large ranks did not produce further gains.
- Tightening the minimum-plays threshold τ produced cleaner vocabularies and improved fit, suggesting that a long tail of low-frequency role-tokens injects more noise than signal at the season sample size we considered.
- The champion-base + role-residual sharing of (6) improved both pick and ban metrics, with the largest gain on ban prediction. This is consistent with bans benefiting most from cross-role parameter sharing, since the ban target is a champion rather than a (champion, role) pair.

What did not help.

- The Plackett–Luce auxiliary loss over future ally picks did not improve held-out performance alone, and only marginally improved top-k accuracy when combined with the bias features at the cost of cross-entropy.
- A standalone ℓ_2 penalty on the strength vector had no effect at small magnitudes and actively harmed ban prediction at larger magnitudes.
- Very small vocabularies (small τ) or very small interaction ranks both hurt fit, consistent with information bottlenecks.

7 Discussion and Future Work

Pro-only data. The model is trained exclusively on professional matches. The professional meta is narrower and more intentional than the broader ranked player base; in particular, certain champions are rarely picked or banned at the professional level and therefore receive imprecise estimates. Augmenting training with ranked data is straightforward in principle but risks contaminating the strength estimate with the substantially different solo-queue meta.

Same-role synergy artifacts. The synergy matrix $\bar{\Sigma}$ is defined on all token pairs, including pairs of tokens at the same role. Such pairs never co-occur as allies in training (a team plays exactly one champion per role), so their $\bar{\Sigma}$ entries are not directly supervised; they are an out-of-distribution extrapolation of the symmetric Gram structure. Diagnostic exports filter same-role pairs by default for this reason.

Static interaction structure. The structure-freezing step assumes that matchup and synergy patterns change more slowly than champion strength. This is a useful inductive bias but is wrong at patch boundaries where item or systemic changes reshape interactions. A natural extension is to allow $\bar{M}$ and $\bar{\Sigma}$ to drift slowly under a small learning rate during the refit phase, rather than treating them as fully frozen.

Downstream uses. The frozen-structure value model is directly usable as a critic for self-play reinforcement learning over the draft phase, as a per-action explanation tool for analyzing past drafts, and as the backend for an interactive draft assistant. Sequence-level losses that score full drafts rather than individual actions are a promising direction for tighter coupling with downstream applications.

Hierarchical pooling. The champion-base / role-residual parameterization is a two-level hierarchy. Extending to additional levels (for example, pooling by champion class or by team composition archetype) could further improve sample efficiency for rare tokens, at the cost of additional design choices about the hierarchy.

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